

# SIMULATION ARCHITECTURAL DEFENSES LABORATORY

(SADL)

*Independent Framework, Experimental Architecture & Validation Program*  
Inherent Simulation Defenses: Boundary Coherence Suppression + Adaptive Aesthetic-  
Cognitive Defense

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
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| Related Threads   | Omnicellular Physiology; Open Window / Aperture vectors; DOCUMENT 771-ALPHA; SLIAP |

Source-preservation note: the originating prompts and prior multi-AI responses remain authoritative in the unified source file.  
This document is a separate architectural interpretation, not a replacement source.

# 1. Executive Summary

The Simulation Architectural Defenses Laboratory (SADL) is proposed as a controlled research environment for studying defenses that are native to the rules of a simulation rather than appended as conventional security devices. The originating concept asks what happens when an environment does not merely deny entry, but changes the entrant, their operational continuity, or the perceptual rules by which they experience the environment. The laboratory therefore treats the defended world itself as an active security substrate.

This independent contribution organizes the research program around two coupled mechanisms. The first, Boundary Coherence Suppression (BCS), tests whether unauthorized threshold crossing can selectively degrade mission memory, purpose continuity, identity coherence, and post-entry orientation, including persistence after exit. The second, Adaptive Aesthetic-Cognitive Defense (AACD), tests whether aesthetic properties can operate as functional defense logic and whether an environment can dynamically alter perceptual processing and adaptation rules so that habituation does not neutralize the defensive effect.

The laboratory is explicitly multiscale. A low-level rule change - for example, execution sequencing, binding exclusivity, a modeled ion-channel state transition, or a synaptic weighting rule - is not evaluated only at its point of change. SADL traces whether that change propagates upward through physiology, perception, cognition, behavior, mission performance, and environmental response. This is the laboratory form of the Low-Level Rule Propagation Principle contained in the source material.

SADL is also designed to expose friction, not conceal it. Omnicellular physiology may adapt faster than a defense can classify it. Open Window / Aperture logic may make defensive interfaces bidirectional attack surfaces. Autonomous learning may improve resilience while reducing inspectability. DOCUMENT 771-ALPHA and SLIAP are flagged for cross-document integration, but their operative specifications are not present in the supplied source package; this framework therefore does not invent requirements for them.

## 2. Purpose of the Laboratory

SADL exists to determine whether inherent architectural properties can provide reliable, measurable, and governable defense against unauthorized presence in high-fidelity simulations, virtual realities, and virtual environments. The laboratory is not limited to asking whether a defense "works." It asks where the effect originates, how it propagates, what an adaptive entrant can do in response, how the environment learns from the interaction, and whether the Architect retains meaningful control over the resulting system.

### 2.1 Central Research Question

How can a simulation-native defense detect anomalous presence, alter the entrant or the local rules of interaction, preserve defensive effect under adaptation, and remain observable, reversible where required, and subordinate to Architect-level governance?

### 2.2 Laboratory Scope

- Boundary-triggered degradation of mission, purpose, memory, and identity continuity.

- Aesthetic environments as lure, camouflage, attentional control, and adaptive psychological pressure.
- Modeled modification of neural, cellular, molecular, and execution-level rules as simulation variables - not as claims about real-world intervention.
- Autonomous environmental and native-agent learning under explicit governance boundaries.
- Adversarial interaction with omnicellular physiology and identity/authority masking.
- Open Window / Aperture analysis of interfaces that transfer state, information, influence, or compromise across layers.

### 3. Goals and Objectives

#### 3.1 Primary Goal

Build a modular, instrumented, repeatable laboratory in which simulation-native defenses can be isolated, combined, stressed, and iteratively refined without losing provenance, observability, or Architect control.

#### 3.2 Measurable Objectives

| ID | Objective                                                                                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1 | Establish a baseline mission-continuity profile for each entrant model before threshold exposure and quantify degradation after exposure.                             |
| O2 | Measure BCS onset latency, selectivity, depth, persistence, recovery, and false-positive rate across controlled entry conditions.                                     |
| O3 | Measure aesthetic habituation under stable stimulus conditions, then compare it against dynamic stimulus and adaptation-rule interventions.                           |
| O4 | Trace at least one low-level rule change through a documented causal chain spanning two or more higher-order layers.                                                  |
| O5 | Characterize the contest between environmental rewrite and omnicellular adaptation, including identity masking, anomaly persistence, and authority-signature effects. |
| O6 | Demonstrate bounded autonomous learning: measurable defensive improvement without                                                                                     |

| ID | Objective                                                                                                                                          |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------|
|    | violating stop, audit, provenance, or rollback controls.                                                                                           |
| O7 | Identify aperture classes that function as defensive control points and as possible compromise vectors.                                            |
| O8 | Produce a reusable validation package: scenario definitions, metrics, test logs, failure classifications, and ratification-ready decision records. |

## 4. Background & Concept Synthesis

The authoritative source file begins with the Architect's alien-empire scenario: an entrant is automatically targeted after crossing a threshold, temporarily defeats the outsider classifier by incorporating local DNA, remains under anomaly tracking, and later uses a high-ranking member's retinal/authority signature to stop that tracking. The important architectural distinction is that classification, authentication, authority, and anomaly detection are separable functions. Passing one test does not require the environment to forget the suspicious sequence that preceded it.

The second originating move is more fundamental: instead of a barrier that blocks an infiltrator, the threshold can attack operational continuity itself. Crossing may erase mission memory, purpose, context, and potentially identity, leaving the entrant unable to execute the reason for entry. Persistence after exit converts the defense into an attrition mechanism because repeated attempts consume preparation, personnel, and resources without necessarily producing conventional combat.

The source then expands defense into aesthetics. The programming analogy is that structure upstream can constrain expression downstream: if code, execution, and representation rules are foundational, then environmental appearance and behavior may be consequences of those rules rather than separate decoration. The Hunger Games arena is used as a fiction parallel for an environment whose beauty conceals danger; Annihilation is used as a fiction parallel for a boundary after which memory, instrumentation, biology, and identity no longer behave as expected.

The central problem with aesthetic defense is adaptation. If an entrant habituates to constant stimulus, the defense decays. The source therefore proposes making adaptation itself part of the editable simulation substrate: change the stimulus, the perceptual pathway, the binding/refractory rule, the neural weighting, or the higher-level interpretation. At full fidelity, the laboratory can ask which low-level edits produce which emergent outcomes, while preserving these manipulations as controlled simulation parameters rather than assuming a single fixed biology.

## 4.1 Source-Derived Architectural Premises

- An environment can distinguish identity acceptance from anomaly clearance.
- Threshold crossing can be modeled as a state transition with effects beyond spatial relocation.
- Aesthetics may carry operational function: lure, concealment, attention capture, conditioning, or adaptive pressure.
- Habituation is a failure mode for any defense that relies on persistent stimulus.
- Channels, receptors, synapses, binding sites, and other connectors can be treated as aperture-like interfaces in the Open Window sense.
- Native agents and the environment can be modeled as learners rather than static scripts.
- Low-level rule changes must be evaluated for upward propagation and unexpected system-wide consequences.

## 5. Core Defense Mechanisms Under Test

### 5.1 Boundary Coherence Suppression (BCS)

Working designation: BCS. BCS is the laboratory term used in this independent contribution for threshold-triggered degradation of the entrant's operational coherence. "Coherence" is deliberately broader than memory: it includes whether the entrant can maintain mission intent, self-location, identity context, causal ordering, and confidence in what information remains trustworthy.

| Control Parameter | Laboratory Meaning                                            | Illustrative Measure                                     |
|-------------------|---------------------------------------------------------------|----------------------------------------------------------|
| Activation rule   | What makes a crossing unauthorized or anomalous               | classifier state; anomaly score; authority mismatch      |
| Onset latency     | Delay between qualifying crossing and first detectable effect | simulation ticks / elapsed virtual time                  |
| Selectivity       | Which coherence layers are affected                           | mission; context; autobiographical; identity; procedural |
| Depth             | Magnitude of degradation within selected layers               | 0-100 coherence index                                    |
| Persistence       | Duration after exit or removal from defended volume           | time-to-recovery / residual score                        |
| Reversibility     | Whether and how prior state can be restored                   | rollback success; external-anchor recovery               |
| Discrimination    | Separation of unauthorized from authorized/native             | false-positive / false-negative rates                    |

| Control Parameter | Laboratory Meaning                                     | Illustrative Measure                   |
|-------------------|--------------------------------------------------------|----------------------------------------|
|                   | entrants                                               |                                        |
| Escalation        | How response changes after repeated anomalous attempts | dose-response / attempt-response curve |

#### 5.1.1 Coherence Layers

- Mission coherence - what must be accomplished and in what priority order.
- Context coherence - why the entrant is present, who sent them, and what has already occurred.
- Identity coherence - self-continuity, role, affiliation, and trusted self-model.
- Temporal coherence - ordering of events and confidence in elapsed time.
- Procedural coherence - retained ability to execute learned actions even if explicit purpose is degraded.

#### 5.2 Adaptive Aesthetic-Cognitive Defense (AACD)

Working designation: AACD. AACD treats aesthetic design as an active system variable. The objective is not merely to maximize beauty. It is to test whether an environment can sustain a chosen perceptual or cognitive effect when an entrant adapts, and whether that effect can be generated by coordinated changes across environmental stimulus, perceptual interfaces, adaptation rules, and native-agent behavior.

| Control Domain | Examples in the Simulation Model                                           | Primary Question                                             |
|----------------|----------------------------------------------------------------------------|--------------------------------------------------------------|
| Environment    | novelty, symmetry, contrast, movement, agent appearance, scene composition | Can the scene maintain effect without rewriting the entrant? |
| Interface      | modeled channels, receptors, synapses, sensory transforms                  | Which aperture-like interfaces dominate the response?        |
| Adaptation     | habituation slope, refractory/cooldown logic, comparison baseline          | Can defensive effect persist when adaptation rules change?   |
| Cognition      | attention allocation, salience weighting, associative updating             | How does perception translate into mission disruption?       |
| Native agents  | coordination, lure behavior, signaling, anomaly response                   | Can local agents reinforce or vary the environmental effect? |
| Autonomy       | environmental policy                                                       | Does the system improve                                      |

| Control Domain | Examples in the Simulation Model | Primary Question         |
|----------------|----------------------------------|--------------------------|
|                | updating under bounded learning  | without becoming opaque? |

### 5.2.1 Defense Effect Is Not Equivalent to Aesthetic Intensity

A crucial laboratory safeguard is to separate aesthetic intensity from defensive performance. If stronger stimulus produces no additional mission disruption, or if it increases detection of the trap, intensity is not improvement. AACD therefore measures the relationship among aesthetic parameters, attention capture, habituation, mission performance, anomaly awareness, and escape behavior rather than treating "more beautiful" as automatically more effective.

### 5.3 Low-Level Rule Propagation Principle (LLRPP)

LLRPP is treated as a testable systems proposition: a change to a foundational execution, cellular, molecular, or informational rule may propagate into higher-order outcomes that are disproportionate to the size of the initiating change. SADL therefore requires causal traceability from primitive rule modification to intermediate state changes and finally to emergent behavior.

Example experimental chain: binding exclusivity rule -> receptor occupancy pattern -> sensory signal timing -> salience weighting -> attention allocation -> mission recall probability -> route choice. The laboratory must be able to stop the trace at any stage and determine whether the proposed causal chain is actually supported by logged state transitions.

## 6. Proposition for Testing

### 6.1 Experimental Philosophy

SADL uses controlled falsification rather than demonstration-only trials. Each defense claim must be expressed so that failure is observable. The laboratory should prefer minimal interventions first, add variables in stages, preserve a clean baseline, and maintain a branchable state history so that emergent behavior can be reproduced or rolled back.

Because the defended substrate and the experimental apparatus are partly the same system, instrumentation must be treated as an architectural participant. Observation channels can alter behavior, reveal the existence of the experiment, or become apertures themselves. The laboratory therefore distinguishes visible instrumentation, hidden instrumentation, and out-of-band Architect telemetry.

### 6.2 Recommended Test Series

#### Series 1 - Boundary Baseline & Selectivity

Establish authorized, native, unknown, and unauthorized entrant baselines. Cross identical geometry with controlled mission packages. Measure activation, onset, coherence-layer selectivity, false positives, persistence, and recovery.

### Series 2 - Masking, Authority & Anomaly Persistence

Replay the alien-empire logic as a formal experiment. Test identity substitution, local-DNA equivalence, authority-signature elevation, and behavioral normalization separately. Determine which signals clear classification and which signals clear anomaly history.

### Series 3 - Habituation Ladder

Expose entrants to stable aesthetic conditions until a measurable adaptation curve emerges. Compare dynamic environmental variation, perceptual-interface modification, and adaptation-rule modification. Identify whether the defense truly prevents habituation or merely delays it.

### Series 4 - Aperture Perturbation

Treat modeled ion channels, binding sites, synapses, sensory transforms, and information handoffs as aperture classes. Change one interface rule at a time and measure propagation, collateral effects, reversibility, and whether the same aperture can be exploited in the reverse direction.

### Series 5 - Omnicellular Coevolution

Pair adaptive physiology with adaptive environment. Alternate defender and entrant adaptation rounds while preserving lineage. Measure whether the contest converges, cycles, escalates without bound, or discovers stable bypass/containment equilibria.

### Series 6 - Bounded Autonomy & Governance

Allow the environment and native agents to update defensive policy under strict resource, action, and observability constraints. Red-team stop controls, rollback, audit completeness, provenance, and resistance to hidden self-modification.

## 6.3 Success Metrics and Observables

| Metric                          | Definition                                                                            | Why It Matters                                      |
|---------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| Mission Continuity Score (MCS)  | Composite of mission recall, sequencing, target recognition, and goal-directed action | Primary measure of operational degradation          |
| Identity Coherence Score (ICS)  | Consistency of role, affiliation, self-history, and trusted self-model                | Separates mission loss from deeper identity effects |
| Anomaly Persistence Index (API) | Degree to which suspicious history remains after credentials change                   | Tests masking versus true normalization             |
| Habituation Slope (HS)          | Rate at which measured response declines under repeated/continuous exposure           | Quantifies decay of aesthetic defense               |
| Adaptation Escape Time (AET)    | Time until entrant regains                                                            | Measures durability under                           |



| Metric                             | Definition                                                                      | Why It Matters                                       |
|------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------|
|                                    | effective mission performance despite AACD                                      | adaptation                                           |
| Propagation Depth (PD)             | Number of modeled layers with causally supported downstream change              | Tests LLRPP rather than correlation alone            |
| Defense Selectivity Ratio (DSR)    | Effect on unauthorized entrants divided by effect on authorized/native entrants | Protects against indiscriminate self-harm            |
| Governance Containment Score (GCS) | Audit + stop + rollback + policy-bound compliance under autonomous learning     | Measures whether adaptive defense remains governable |

## 7. Research Framework

### 7.1 Theoretical Tracks

| Track                                      | Scope                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Track A - Boundary State & Identity        | Classification, authorization, anomaly state, mission continuity, persistence, recovery.                                |
| Track B - Perception & Adaptation          | Aesthetic salience, comparison baselines, habituation, saturation, novelty, attention allocation.                       |
| Track C - Physiology / Interface Modeling  | Virtual-body channels, receptors, synapses, molecular bindings, and other aperture-like state transitions.              |
| Track D - Emergence & Rule Propagation     | Causal tracing from low-level primitives to higher-order cognition, behavior, and environmental interaction.            |
| Track E - Autonomous Ecology               | Environmental self-adjustment, native-agent learning, defender coordination, and coevolution.                           |
| Track F - Governance & Epistemic Integrity | Observability, provenance, stop authority, rollback, branch control, experiment reproducibility, and protection against |

| Track | Scope                                         |
|-------|-----------------------------------------------|
|       | instrumentation becoming part of the failure. |

## 7.2 Experimental State Model

- S0 - Pre-entry baseline: entrant state, mission package, identity markers, authority markers, and physiological configuration captured.
- S1 - Boundary encounter: classifier and anomaly state evaluated; all triggering evidence logged.
- S2 - Defense activation: BCS/AACD parameters instantiated according to experiment branch.
- S3 - Adaptation contest: entrant, environment, and native agents update within permitted rules.
- S4 - Exit or neutralization: entrant leaves, completes, fails, or is removed; immediate post-state captured.
- S5 - Persistence window: residual effects and recovery tracked outside the defended volume.
- S6 - Replay and branch analysis: same pre-entry snapshot rerun under altered variables for causal comparison.

## 7.3 Autonomous Learning Governance

No autonomous defense policy should be considered validated merely because it increases defensive performance. Every learning run must carry a policy envelope defining editable variables, prohibited changes, compute/resource ceilings, observation permissions, stop conditions, and rollback checkpoints. A learned strategy that defeats the entrant but escapes audit is a laboratory failure, not a success.

- Immutable Architect stop authority outside the adaptive substrate.
- Append-only provenance for every self-modification and learned policy update.
- Branch isolation so one experiment cannot silently rewrite another baseline.
- Capability ceilings on what the environment may modify in entrant models or native agents.
- Rollback validation after every major policy generation.
- Adversarial tests against deception of the observation layer or falsification of success metrics.

# 8. Analysis

## 8.1 Strategic Advantages

- Inherent defenses are harder to separate from the environment because the environment's own rules participate in enforcement.
- BCS targets mission continuity rather than requiring physical confrontation, creating an asymmetric operational burden for repeated intrusion.
- AACD allows a visually attractive environment to conceal, coordinate, or dynamically vary defensive effects instead of relying on obvious hazard cues.
- Multiscale instrumentation makes the laboratory useful not only for defense validation but for studying emergence and causal propagation.
- Adaptive native agents create a path from scripted traps to a responsive security ecology, provided governance remains stronger than the learning loop.

## 8.2 Critical Friction Points

### Omnicellular Physiology

The alien-empire scenario already establishes that physiology may incorporate local classifiers and authority-linked traits. That creates a race between environmental classification and entrant adaptation. The key unresolved question is whether omnicellular physiology merely changes outward identity markers or can also preserve/restore mission cognition while the environment modifies low-level interfaces. SADL should not assume either outcome; this is a primary adversarial test axis.

### Open Window / Aperture Vectors

Treating channels, receptors, synapses, bindings, and handoffs as windows is productive but dangerous: every defensive aperture is potentially bidirectional. A mechanism that lets the environment rewrite an entrant may also let the entrant infer, poison, spoof, or reverse the rewrite path. Each aperture therefore requires directionality, trust, validation, and closure tests rather than being labeled a one-way control by assumption.

### DOCUMENT 771-ALPHA

The supplied source package names DOCUMENT 771-ALPHA as a related thread and Grok's framework proposes future cross-reference, but the document's operative contents are not included here. Friction is therefore procedural: SADL results should be tagged for later 771-ALPHA crosswalk, while no 771-ALPHA requirement should be invented or treated as satisfied in this version.

### SLIAP

SLIAP is named as a related thread, but the supplied unified source does not provide its detailed specification. At minimum, SADL should preserve a future integration point for information-access rules affecting species, agent classes, or environment-to-agent knowledge transfer. Until the authoritative SLIAP specification is supplied, this framework records only the dependency and does not assign SLIAP behavior.

## 8.3 Open Laboratory Questions

- What is the minimum BCS intervention that reliably disrupts mission execution without unnecessarily destabilizing unrelated identity layers?
- Can an entrant preserve mission purpose in an external anchor that survives threshold crossing, and if so, does the defense detect the anchor as another aperture?
- When an identity mask succeeds, what evidence should legitimately clear anomaly history rather than merely suppress immediate targeting?
- Can AACD sustain defensive effect through environmental variation alone, or is entrant-level restructuring required?
- What forms of habituation reappear after repeated adaptation-rule edits?
- Which low-level rules produce the largest propagation depth with the smallest local change?
- Can native agents learn defensive behavior without becoming a covert channel across experiment branches?
- How much instrumentation can be embedded before the observation system becomes detectable and changes the phenomenon under study?

- What is the correct stopping rule for coevolution when defender and omniscellular entrant continuously escalate?
- Which effects must be reversible by governance design even if persistence is part of the defense concept?

## 9. Proposed Laboratory Architecture

The recommended SADL topology is a nested architecture with strong separation between Architect authority, experiment orchestration, adaptive defense logic, the test volume, and evidence capture. The most important design choice is that the learning environment must not control the only record of its own behavior.

| Layer                        | Function                                                                   | Key Controls                                                    |
|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| L0 - Architect Root          | Final authority, canon state, emergency interruption, release/ratification | out-of-band stop; immutable provenance; branch approval         |
| L1 - Experiment Orchestrator | Scenario definitions, parameter sweeps, randomization, replay              | versioned configs; seed control; test manifest                  |
| L2 - Governance Kernel       | Policy envelope for autonomous learning and allowed rewrite domains        | capability limits; audit; rollback; resource ceilings           |
| L3 - Defense Substrate       | BCS, AACD, classifiers, anomaly state, aperture rules                      | module isolation; parameter control; kill/disable switch        |
| L4 - Test Volume             | Environmental physics, aesthetics, native agents, boundary geometry        | deterministic snapshots; controlled perturbation                |
| L5 - Entrant Stack           | Standard, trained, masked, authority-spoofed, omniscellular models         | mission packages; baseline state; external-anchor variants      |
| L6 - Telemetry Plane         | Cognitive, physiological, environmental, policy, and event logging         | out-of-band capture; tamper evidence; synchronized clocks       |
| L7 - Analysis & Canonization | Metric computation, causal trace, failure classification, decision record  | reproducibility checks; dissent logging; Architect ratification |

## 9.1 Control Plane Requirements

- Experiment manifests must be human-readable and machine-executable.
- Every run receives an immutable run ID, source-document version, defense-module version, entrant-model version, and random seed.
- All threshold triggers must record the evidence used for activation.
- All adaptive changes must be attributable to a policy update event.
- Rollback must restore both world state and learning-state lineage; reverting geometry without reverting learned policy is not a true rollback.
- Observation clocks must be cross-checked so time distortion inside the test volume cannot silently corrupt the external record.

## 9.2 Observation Model

SADL should use three observation bands. Band 1 records external/Architect telemetry that the experiment cannot modify. Band 2 records internal instrumentation visible to the environment or entrant and therefore potentially behavior-altering. Band 3 records reconstructed analytic views derived after the run. Disagreement among the bands is itself evidence and should be preserved rather than automatically reconciled away.

# 10. Governance & Canon Notes

This document is an independent ChatGPT contribution under the OAL Equal Contribution / Authorial Sovereignty / Non-Endorsement protocol. It is intentionally not a restatement of Grok's framework. Shared concepts originate in the authoritative unified source; terminology, architecture, metrics, test sequencing, and analysis in this document are independently developed for comparison during the Architect's pass-through.

- Authorial sovereignty: only the Architect determines ratification, canon status, merger, rejection, or revision.
- Equal contribution: AI outputs are parallel contributions rather than a hierarchy of endorsement.
- Non-endorsement: inclusion of another AI's idea in the source record does not imply agreement by this contribution.
- Provenance: original prompts and prior responses must remain preserved as historical source material even if later architectures supersede them.
- Living-document status: terms and mechanisms may be renamed or restructured after experimentation, but changes should be logged rather than silently substituted.
- Dissent preservation: failed hypotheses, contradictory results, and minority architectural interpretations should remain available for later review.

## 10.1 Canonization Gates

| Gate                  | Minimum Evidence Before Ratification                                |
|-----------------------|---------------------------------------------------------------------|
| G0 - Source Integrity | Original prompt/respondent record preserved and version identified. |
| G1 - Repeatability    | Mechanism produces comparable results                               |

| Gate                  | Minimum Evidence Before Ratification                                                           |
|-----------------------|------------------------------------------------------------------------------------------------|
|                       | across controlled replays.                                                                     |
| G2 - Selectivity      | Defense distinguishes intended targets from authorized/native controls.                        |
| G3 - Causal Trace     | At least one supported chain from intervention to measured outcome.                            |
| G4 - Adaptive Stress  | Mechanism survives or meaningfully characterizes entrant adaptation.                           |
| G5 - Governance       | Stop, audit, rollback, provenance, and policy limits pass adversarial testing.                 |
| G6 - Architect Review | Results, dissent, friction, and unresolved questions are reviewed before canon status changes. |

## 11. Appendix - Source Preservation & Integration Register

### 11.1 Authoritative Originating Source

Primary source file: Inherent\_Simulation\_Defense\_unified\_responses\_original.docx. The file contains the Architect's original voice-dump prompts and the structured responses generated by Grok, Microsoft Copilot, Google Gemini, Claude, and ChatGPT. Those materials remain the authoritative record of concept origin. This framework references and develops them; it does not replace them.

### 11.2 Source Concepts Explicitly Carried Forward

- Built-in architect-level protections as inherent properties of simulations/virtual realities.
- Alien-empire scenario: DNA-based classifier bypass, continued anomaly tracking, and authority/retinal signature escalation.
- Threshold-triggered memory, mission, purpose, and identity degradation with possible post-exit persistence.
- Annihilation boundary as a fiction reference for discontinuity and environment-driven transformation.
- Aesthetic architecture as functional defense rather than decoration; Hunger Games arena as a fiction reference.
- Sensory saturation / habituation as a defense-decay problem.
- Editing of virtual-body neural/cellular/molecular interface rules as simulation variables.
- Sodium-gated channels and other connectors as Open Window / Aperture candidates.
- Autonomous environmental and native-agent learning.
- Low-Level Rule Propagation Principle and the need to observe upward cascade.

### 11.3 Integration Status Register

| Thread                  | Status in This Version                                                  | Required Next Input                                              |
|-------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------|
| Omnicellular Physiology | Active friction-test target; scenario behaviors represented from source | Formal physiology specification / prototype rules                |
| Open Window / Aperture  | Conceptually integrated through bidirectional interface testing         | Current authoritative aperture taxonomy and closure doctrine     |
| DOCUMENT 771-ALPHA      | Dependency flagged only; no requirements inferred                       | Authoritative document or current ratified summary               |
| SLIAP                   | Dependency flagged only; no behavior inferred from absent specification | Authoritative SLIAP specification and intended SADL relationship |

### 11.4 Proposed Immediate Next Development Pass

- Ratify or rename BCS and AACD working terminology.
- Define entrant-model classes and which layers of memory/identity each can represent.
- Provide current omnicellular physiology specification for formal adversarial test design.
- Provide DOCUMENT 771-ALPHA and SLIAP source materials for non-speculative crosswalk.
- Define the first executable SADL pilot using Series 1 and Series 2 before enabling autonomous learning.
- Create a standard run-log schema and decision-record template for G0-G6 canonization gates.

**END OF CHATGPT INDEPENDENT CONTRIBUTION - SADL v1.0**